

Demand Sensing Intelligence

A defensible research priority,
with evidence behind every score

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Agentic workflow / Internal retrieval / Research decisions

The research decision and the case-study brief

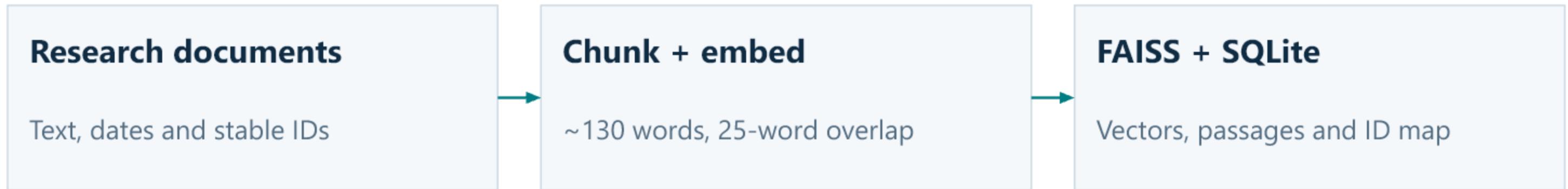
Where should we commission, refresh or maintain research?

Research planning	Client question answering
Public signals reveal emerging themes	Internal retrieval finds relevant passages
Coverage and gaps rank possible work	Citations connect claims to research notes
A knowledge graph exposes relationships	Support review flags unsupported claims

LangGraph coordinates both routes. The simulated library defines the boundary of what the assistant knows.

Architecture: setup and a live request

SETUP: PREPARE THE INTERNAL LIBRARY

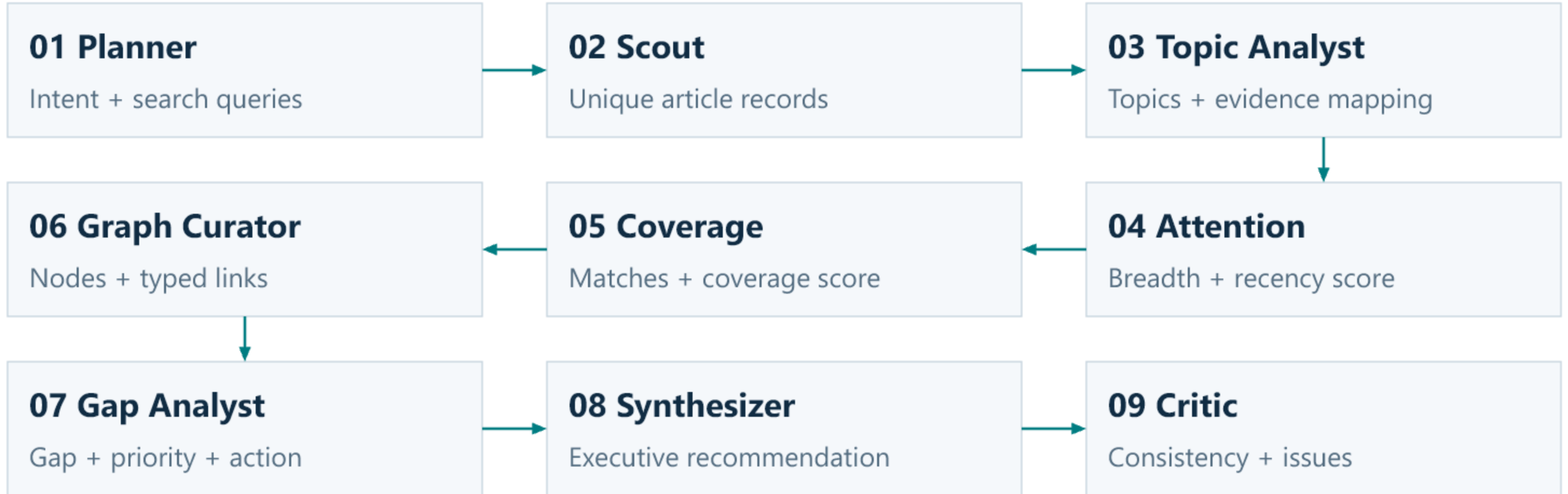


INFERENCE: ONE QUESTION FROM THE UI



Inference reuses the index. Tavily supplies public signals, Gemini supplies model calls, NetworkX builds relationships.

The market-scan workflow



Shared state carries the original question, articles, topics, coverage, graph, scores and final narrative.

Planner and Scout: a question becomes evidence

UI input: “AI in healthcare” + Market scan

Planner output: queries[]	Scout action
AI in healthcare	Tavily: request up to 8 results
AI in healthcare emerging trends	Tavily: request up to 8 results
AI in healthcare regulation governance	Tavily: request up to 8 results

Pool results, deduplicate URLs, retain article metadata

Output: articles[] + signal_source + scout_stats. Three queries can return fewer than 24 unique articles.

Topic extraction: what the first model call reads

INPUT: SIX ARTICLES PER CALL

[0] Hospital A establishes an AI oversight committee
Date: 2026-09-13
Source: hospital.example
Excerpt: committee reviews models before deployment...

OUTPUT FOR ARTICLE INDEX 0

topics: ["Clinical AI Governance"]
entities: [{name: "Hospital A", type: "ORGANISATION"}]
driver: "New committee reviews clinical models"

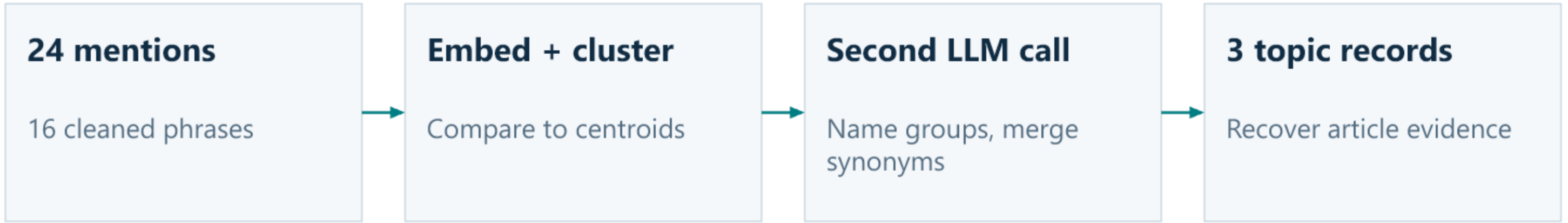
12 articles × 2 themes = 24 extracted mentions

Clean phrase keys and retain their article indexes.

Example: "The AI governance trends" becomes "ai governance".

Illustrative extraction. The model returns themes, entities and a driver. Code creates graph relationships later.

Topic consolidation: group, name, map back



Similarity = dot(unit vector, unit centroid) Join at ≥ 0.93

Canonical output example	Member groups	Recovered evidence
Clinical AI Governance	[0, 2]	5 article URLs across 4 domains
Healthcare Agentic AI	[1]	Its own URLs, entities and dates
AI Diagnostic Accuracy	[3]	Its own URLs, entities and dates

Illustrative counts. Output is 3 topic records plus the original articles in state. Topic names emerge from this scan.

Attention: breadth and recency, calculated explicitly

Example input: 5 articles, 4 publishers, newest article 4 days ago

BREADTH: REWARD DISTINCT PUBLISHERS

$$B = \min(1, 0.7 \times \text{publishers} / 6 + 0.3 \times \min(\text{articles}, 12) / 12)$$

$$B = 0.7 \times 4/6 + 0.3 \times 5/12 = 0.5917$$

RECENCY: DISCOUNT OLDER SIGNALS

$$R = \exp(-\text{days} / 30) = \exp(-4 / 30) = 0.8752$$

$$\text{Attention} = 0.7B + 0.3R$$

0.6767

The state field is momentum. This score measures sampled attention. Historical growth is not measured.

Coverage: the retrieval funnel for each topic

Query = topic label + description



Python filters candidates

Exclude Syndicated Archive.
Require similarity ≥ 0.62 .
Several chunks from one note count as one document.

Loop over the topic list

3 topics mean 3 coverage assessments.
Each topic receives its own matches, reasons, coverage score and staleness.

The judge reads excerpts and returns reasons

INPUT TO THE MODEL

Clinical AI Governance: oversight and accountability for clinical models.

For each candidate: document ID, title, metadata, similarity and a 420-character excerpt.

Illustrative candidate excerpt	Model verdict	Reason returned
RN-9001: approval gates and clinical AI oversight	covers	Directly addresses governance
RN-9002: an audit subsection on clinical models	partial	Covers one part of governance
RN-9003: diagnostic accuracy benchmarks	tangential	Same domain, different subject

Output: [{doc_id, verdict, reason}]. Python assigns weights of 1, 0.5, 0, 0 and calculates the scores next.

Coverage combines relevance, age and depth

Freshness $F = 1$ through day 120, 0 from day 450

Between them: $F = 1 - (\text{age} - 120) / 330$ At 285 days: $F = 0.50$

Usable weight $u = \text{relevance} \times (0.25 + 0.75 \times F)$

Accepted document	Verdict weight	Age	F	Usable u
RN-9001: covers	1.0	500 days	0	0.250
RN-9002: partial	0.5	500 days	0	0.125

$C = 0.6 \times \min(1, \sum u / 5) + 0.4 \times \max(u)$

0.145

$C = 0.6 \times (0.375 / 5) + 0.4 \times 0.25 = 0.145$. Staleness: 500 days, the age of the newest accepted note.

The graph connects a topic to its evidence



Input is already in shared state

Topic entities from article extraction.
Accepted internal documents from Coverage.
Code assigns the relationship types.

Output belongs to the selected run

Nodes, links, graph statistics and central entities.
The UI shows the map for that question.
SQLite preserves the result for replay.

Illustrative relationship. "Covered by" may mean partial or stale coverage. The graph does not recalculate the gap.

Gap scoring turns the evidence into a priority

$$\text{Gap} = \text{attention} \times (1 - \text{coverage})$$

$$0.6767 \times (1 - 0.145) = 0.5786$$

Priority: CRITICAL

Gap ≥ 0.45
Strong sampled attention meets weak usable coverage.

Action: REFRESH

Attention ≥ 0.50 , coverage < 0.70
Newest accepted note > 300 days old
Refresh rule takes precedence over commissioning.

Same worked example. Output: gap_score 0.5786, priority CRITICAL, quadrant refresh and an evidence-based rationale.

Synthesis writes the decision; Critic checks the numbers

SYNTHESIZER INPUT

Clinical AI Governance: attention 0.6767, coverage 0.145, gap 0.5786.
CRITICAL / refresh. Newest accepted note: 500 days old.

EXECUTIVE OUTPUT

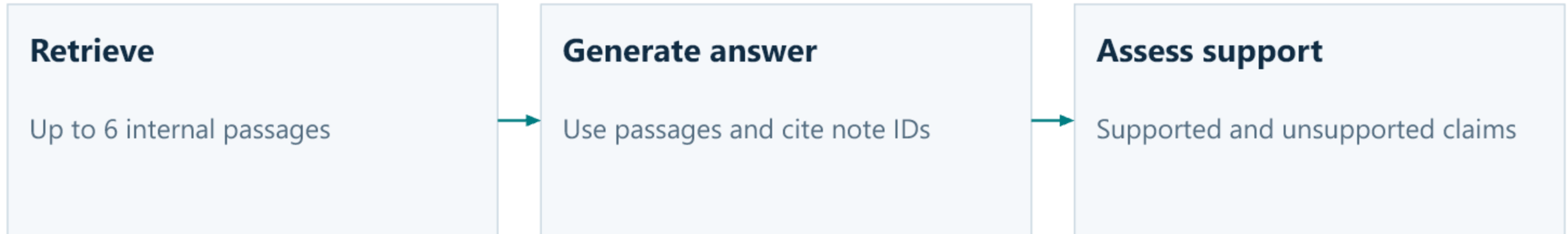
“Refresh the clinical AI governance research. Current coverage is dated, while this news sample shows substantial attention to the topic.”

Critic compares the summary with the supplied analytics.

Illustrative summary. Output adds the narrative, computed actions, consistency verdict and any flagged issues.

Library QA answers a different user need

“What governance practices does our research recommend for clinical AI?”



Output: answer + citations + retrieved evidence + support assessment

No relevant evidence: abstain. Library QA does not need a web scan, attention score or gap calculation.

Demonstration evidence and pilot KPIs

Recorded example: 23 articles / 9 topics / ~64 seconds

Already observable	Pilot measurement to add
Saved question, agent trace and graph snapshot	Replay reliability across representative questions
Latency, calls and token counts	p50/p95 latency and cost per successful run
Retrieved passages and judge reasons	Recall@k, citation precision and analyst agreement
Ranked actions and model consistency check	Recommendation acceptance and analyst time saved

Historical run #9 used cached search results. Operational success does not establish production accuracy or business impact.

The next investment is stronger evidence

Current boundary	Next investment	Success measure
Heuristic scores and model verdicts	Analyst-labelled cases and calibration	Agreement on useful research actions
Short excerpts, one passage per note	Richer evidence and retrieval evaluation	Fewer missed and false coverage matches
News sample without a time baseline	Comparable historical signal collection	Measured change over time
Local prototype and shared services	Access control, durable jobs, telemetry	Reliable use under concurrent load

A controlled analyst pilot should establish quality, usefulness and operating cost before wider deployment.

A research decision with a visible calculation

Clinical AI Governance

Attention 0.6767
Coverage 0.145
Gap 0.5786

REFRESH

Two accepted notes,
both 500 days old

The analyst sees the evidence and owns the final decision.

The dictionaries preserve provenance during cleanup

Cleaned key	surface_counter	surface_articles	surface_original
ai governance	3	[0, 1, 2]	AI Governance
agentic ai	1	[0]	Agentic AI
clinical ai oversight	1	[2]	Clinical AI Oversight

Merged topic evidence: [0, 1, 2] + [2] gives 3 unique article URLs

The maps store phrase frequency, provenance and display spelling. Deduplicating a phrase does not discard its source articles.

Centroid comparison and the second model call

$v_0 = (1, 0)$ starts group 0

$v_1 = (0.96, 0.28)$: $\text{dot}(v_1, v_0) = 0.96$, so join

New centroid = $\text{normalize}(\text{mean}(v_0, v_1)) \approx (0.98995, 0.14142)$

SECOND CALL INPUT: PHRASES + GROUP IDs

[0] Clinical AI Governance

[1] Healthcare Agentic AI

[2] Clinical AI Oversight

SECOND CALL OUTPUT: TOPIC + MEMBERS

Clinical AI Governance: member_groups [0, 2]

Healthcare Agentic AI: member_groups [1]

Each also returns category and description.

member_groups recovers phrases, then article indexes, then URLs and entities. The naming call receives no raw vectors.

Agent outputs accumulate in shared state

Node	Main fields it adds or updates
Planner / Scout	intent, research_area, queries / articles, signal_source, scout_stats
Topic / Attention	topics, topic_objects, normalisation / momentum, recency_days
Coverage / Graph	coverage[slug], matched_docs / graph.snapshot nodes and links
Gap Analyst	gaps: scores, priority, quadrant, rationale / portfolio summary
Synthesizer / Critic	answer, recommendations / verdict, groundedness, critique_issues

Each node reads the shared state and returns a partial update. Topic and coverage records join through the topic slug.

Score interpretation and failure boundaries

Condition	What it means in this implementation
1 fresh “covers” document	Coverage = $0.6 \times (1/5) + 0.4 \times 1 = 0.52$
5 fresh “covers” documents	Coverage = 1.00, the depth target is met
Judge unavailable or incomplete	Labelled similarity proxy: partial only at similarity ≥ 0.74
No accepted documents / missing dates	999 is a sentinel. No accepted docs also gives coverage 0.
Index unavailable	Coverage unavailable. Skip gap scoring for that topic.

Staleness is calendar age of the newest accepted note. A model verdict, heuristic score and confidence estimate are different things.